

Participant 33 — T0 method (label-free)

Theory behind `Results_33_T0_0.csv` (PPP_WTG17) and `Results_33_T0_final.csv` (SSS_WTG06). Both files come from the same model with no fitted parameter:

$$\hat{\theta}(d) = L - (\bar{r}_{\ell(d)} - \alpha_{\text{seg}(\ell(d))})$$

Symbol	Meaning	Source
$\hat{\theta}(d)$	predicted static yaw misalignment on day d	
$\ell(d)$	the day's state; submitted as <code>cluster</code>	change points of r (§3)
$\bar{r}_{\ell}$	level of the neighbour heading residual in state ℓ	§2
α_{seg}	anchor: day-weighted mean of $\bar{r}_{\ell}$ over an encoder-free segment	§4
L	level prior: the turbine's vane set-point	§4

In words: the *changes* of the misalignment are read from how the nacelle heading moves against its neighbours, with a slope of exactly -1 that follows from the sensor algebra; the *level* is taken from the vane set-point.

1. Sensor model

Unobserved: true wind direction W , true nacelle heading N , yaw error $\gamma = W - N$ (wrapped to $\pm 180^\circ$). The SCADA stream reports

$$N_r = N + \delta_n, \quad v = g\gamma + \delta_v, \quad W_r = N_r + v.$$

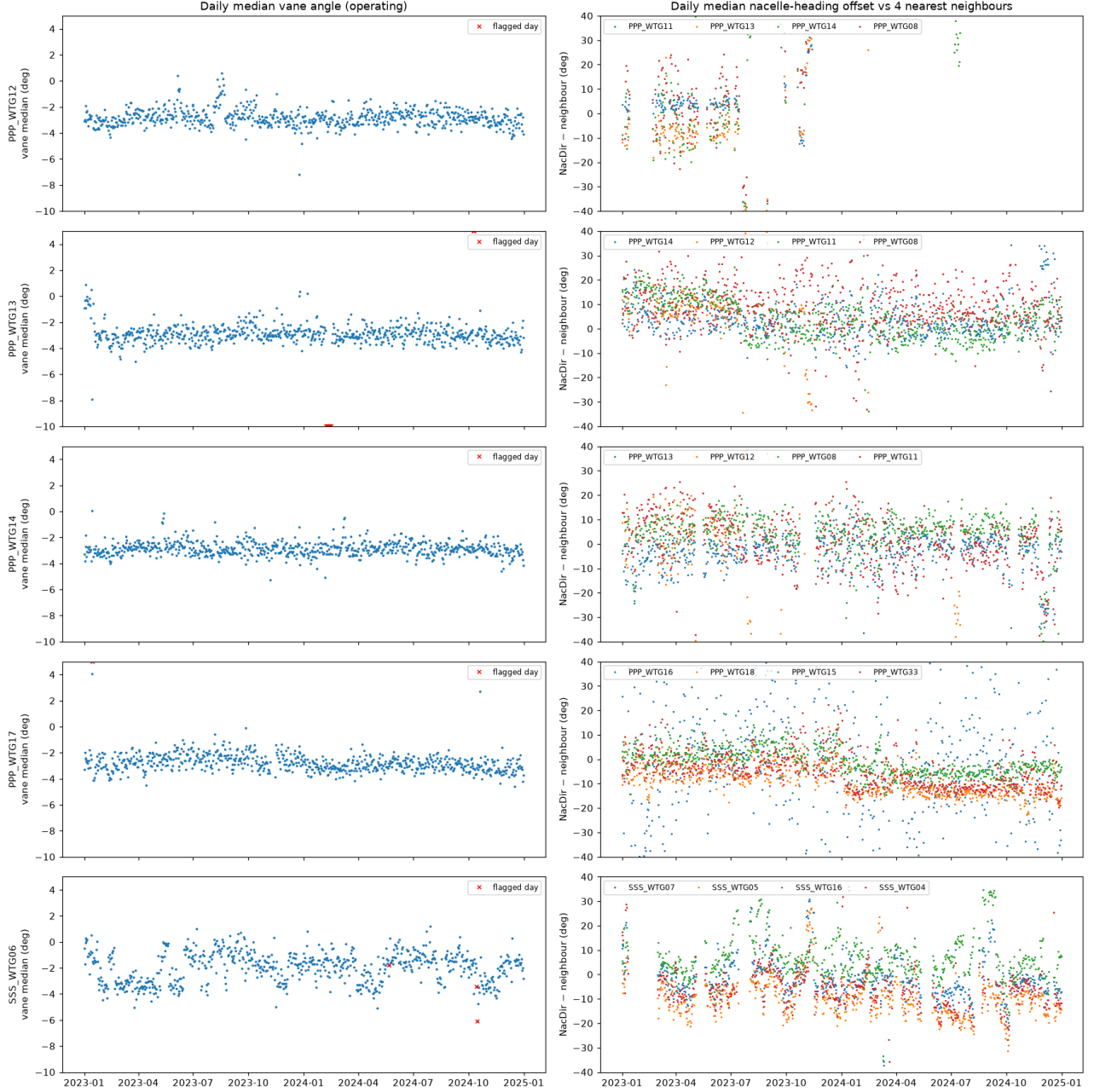
- δ_n : the nacelle encoder has no north reference. It is an arbitrary constant that jumps whenever the encoder is re-referenced.
- v : the vane reads the yaw error with a mounting offset δ_v and a gain $g \leq 1$ (we use $g = 1$; it cannot be measured from this data).
- The third equation is how the data set is encoded, not an assumption: the vane is recovered as `vane = wrap(WindDir - NacDir)`, so `WindDir` carries the encoder offset and only the vane is encoder-free.

The yaw controller rotates the nacelle until the filtered vane sits at its set-point s , so in steady state $E[v] = s$ and

$$\theta \equiv E[\gamma] = \frac{s - \delta_v}{g}.$$

θ is the static misalignment: the persistent yaw error the controller cannot see, because it sits inside its own measurement.

Consequence: the vane cannot see θ . Averaged over a day the vane reads s whatever θ is. The left column below shows it: the daily median vane is flat at about -3° on every PPP turbine. Information about θ has to come from an external direction reference, and the neighbours provide one (right column: heading differences against the four nearest turbines, before any cleaning).



2. The neighbour heading residual

Let $c = W + \kappa$ be a consensus of the wind direction from the neighbours, with κ a constant made of the neighbours' own offsets. Then

$$N_r - c = N + \delta_n - W - \kappa = -\gamma + \delta_n - \kappa,$$

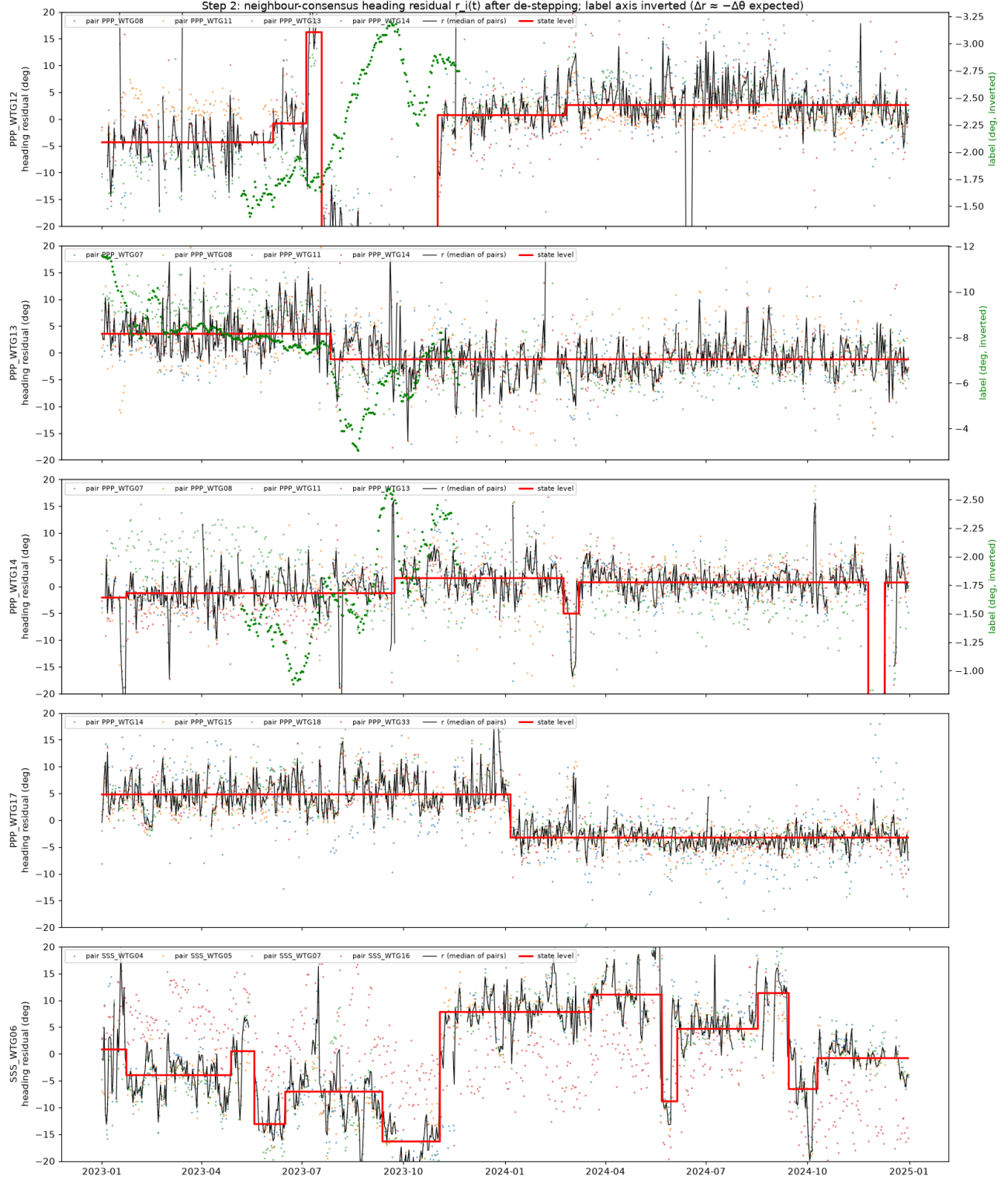
and averaged over a day

$$r(d) = -\theta(d) + \delta_n(d) - \kappa \quad \implies \quad \Delta r = -\Delta\theta \quad \text{whenever } \Delta\delta_n = 0.$$

So the residual gives **changes** of the misalignment in physical degrees, with slope -1 , independent of the vane and of any power model. Its **level** is not identifiable, because δ_n and κ are unknown constants.

How r is built (10-minute aggregates, circular statistics throughout):

1. **Pair series.** For turbine i and each of its 4 nearest same-site neighbours j : $D_{ij} = \text{wrap}(N_r^i - N_r^j)$, in bins where both turbines operate.
2. **Encoder de-stepping.** A re-reference of i moves D_{ij} against *all* neighbours by the same amount on the same day. Jumps of at least 25° with the same sign in at least 60 % of the pairs within ± 3 days are credited to the encoder and subtracted.
3. **Excursion masking.** Days more than 30° from the pair's 31-day rolling median are dropped.
4. **Sector correction.** Terrain and wakes make the heading difference depend on wind direction, so the median of D_{ij} per 30° sector of the *neighbour's* wind direction is subtracted. This absorbs the neighbour's own offsets into κ .
5. **Robust daily value.** Median over the bins of the day per pair, then median over pairs: $r_i(d)$. Day-to-day noise is about 2° .



Coloured dots are the individual pairs, the black line is r , the red line the state levels of §3. On the three training turbines the label is drawn in green on an inverted axis, so r and the label should move together.

3. States

r is smoothed with a centred 7-day rolling median and segmented with PELT (squared-error cost, penalty 400, minimum state length 14 days). The penalty means a jump J lasting n days on both sides is accepted when $nJ^2/2 \gtrsim 400$, e.g. 2.4° over 140 days or 5° over 30 days. Each state gets the median of r as its level $\bar{r}_\ell$.

A boundary with a jump above 15° is typed as an **encoder** boundary: too large for a misalignment change, and step 2 only removes re-references of 25° or more. r is not comparable across such a boundary.

The states are computed from the SCADA residual over all 731 days, with no knowledge of the template or of which days are scored. They are the **cluster** column.

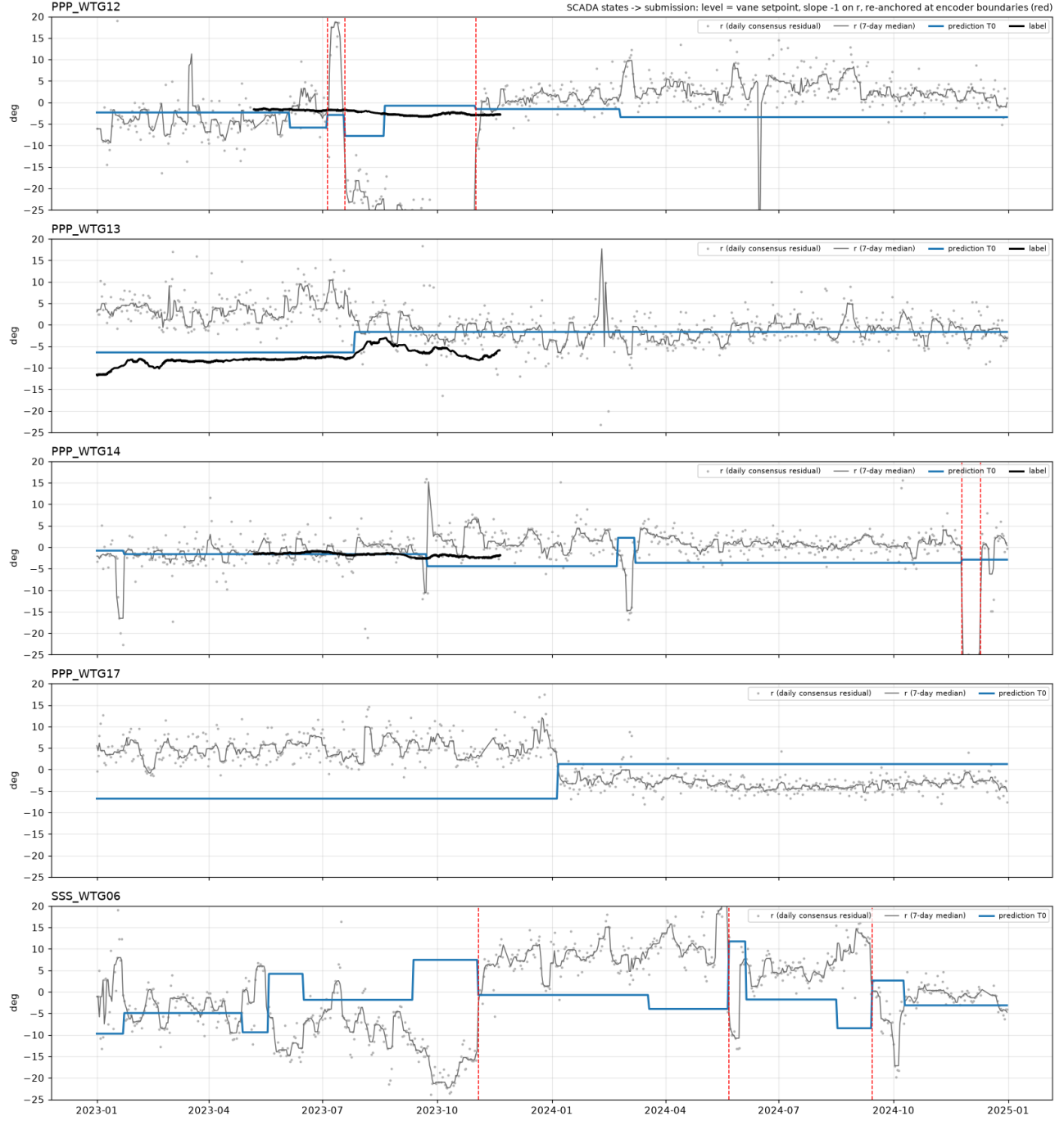
4. Level: vane set-point and anchor

No label-free estimator of the absolute level survived testing. Locating the peak of the power-versus-yaw curve fails because over the controller's $\pm 8^\circ$ deadband the power changes by only about 2 % ($\cos^2$ model), the same order as sector, turbulence and season effects.

The level is therefore a prior. Assuming an unbiased vane ($\delta_v = 0$, $g = 1$), $\theta = s$, and s is measured as

$$L = \text{median}_d \text{ healthy } (\text{daily median vane in operation}).$$

The prior describes the turbine's *typical* state, so the state levels are centred before the slope is applied. Within each encoder-free segment (a run of states with no encoder boundary) the anchor α is the day-weighted mean of the state levels. Re-anchoring per segment stops an undetected re-reference from leaking into the predictions.



Blue is the prediction, grey the residual it is derived from, black the label where it exists (training turbines only). Red dashed lines are encoder boundaries.

5. What was submitted

File	Turbine	States	Predicted levels
Results_33_T0_0.csv	PPP_WTG17	2 (boundary 2024-01-06, $\Delta r = -8.05^\circ$)	-6.8° then $+1.3^\circ$
Results_33_T0_final.csv	SSS_WTG06	13, in 4 encoder-free segments	$-9.7^\circ \dots +11.8^\circ$, mean -1.8°

6. Use of labels

No parameter of the model is fitted to labels: the slope is -1 from §2 and the level is the vane set-point. The training labels were used for two things only.

- **Sign check.** The script refuses to write a file unless the label moves opposite to r on the training turbines (measured slope -0.74), which fixes the sign convention of θ .
- **Evaluation.** Scored on the three training turbines with the challenge rule, the model reaches RMSE 2.48° , MAE 1.96° , bias $+0.54^\circ$, ARI 0.54. Segmentation thresholds were set while these turbines, with their labels, were in view.

7. Known limits

- **The level is a prior, not a measurement.** It is wrong by $-\delta_v$ when the vane itself is offset. On the training turbines it is within 1° on two and 4.5° off on the third.
- **A small re-reference looks like a misalignment change.** Below 15° the two cannot be told apart in r , so boundaries are more reliable than magnitudes.
- **Steps below about 1° are under the noise floor of r .**
- **SSS_WTG06 is the uncertain case.** The residual implies swings of -10° to $+12^\circ$ across 13 states. If part of that is site-specific drift of r rather than misalignment, the final file over-predicts the spread.